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Probability

PROB101 | Suggested sequence · Semester 1 | Student-authored study guide

How to use these notes

Events, conditional probability, Bayes' rule, random variables, common distributions, and expectation. Work through the units in order, reproduce each example, then answer the review prompts from memory. This is supplemental practice material, not an official MBU syllabus or assessment. Course sequencing is suggested and should be checked against your current approved syllabus.

MODULE 01 · Events and conditional probability

Probability ranges from 0 to 1. For mutually exclusive events, P(A"*B)=P(A)+P(B); in general subtract the intersection. Conditional probability is P(A|B)=P(A")B)/P(B) when P(B)>0.

Worked example

```
If 30 of 100 students take both subjects and 50 take math, P(data|math)=30/50=0.6.
```

Check your understanding

- When are two events independent?
- How does conditional probability change the sample space?

MODULE 02 · Bayes' rule and random variables

Bayes' rule reverses conditioning: P(A|B)=P(B|A)P(A)/P(B). A random variable maps outcomes to numbers. Its distribution describes probabilities; expectation is a weighted average, not necessarily a possible outcome.

Worked example

```
With a rare condition, a positive test can still have a modest posterior probability if false
positives are common.
```

Check your understanding

- Why does prevalence matter in a diagnostic test?
- What is expected value?

MODULE 03 · Distributions and limit behavior

Bernoulli models one binary trial; binomial counts successes in fixed independent trials; normal distributions model continuous values with a symmetric bell shape. The central limit theorem concerns sampling distributions under conditions, not raw data always becoming normal.

Worked example

```
For n independent Bernoulli trials with success probability p, the count has mean np and
variance np(1-p).
```

Check your understanding

- What assumptions define a binomial model?

- What does the central limit theorem describe?

Revision checklist

- %¡ Can I explain the main idea without copying the notes?
- %¡ Can I reproduce the example and state its assumptions?
- %¡ Can I test an edge case and describe what should happen?
- %¡ Have I recorded a limitation and checked a primary reference?

Further reading

Use the official documentation and open references linked in the portal. Check the publication/version and license at the source before redistribution. Cite the source and access date in coursework.